

Maha Strategies LLC

Infrastructure Report — Second Assessment

Prepared 3 August 2026 | Follows the first assessment of 2 August | main at the merge of PR #11 | All figures measured against Production during preparation

The first assessment found a well-built platform with a thin operational envelope: schema changes applied by hand, no verified backup, and alerting that ended in one inbox. Since then fourteen commits have shipped across three pull requests. Most of that envelope now exists and is running in Production. Two of the three original P1 findings are closed or materially reduced; the third — escalation — proved itself real during the work when release health failed for over an hour and nothing told anyone. The measurement layer the machine-economy strategy depends on is now live, and it has already found a revenue line that cannot transact.

265

Tests passing

59

DB migrations

94

API routes

10

CI workflows

0

Vulnerabilities

1. What was verified, and how

Every line below was checked against Production or the live GitHub API during preparation, not inferred from the repository.

Check	Result
Release health (was failing at last report)	Recovered — successive green runs; last-known-good artifact producing again
Root cause of that failure	RELEASE_HEALTH_TOKEN absent from Vercel Production; auth fell through to the revenue token
Signed operations alerts	Receiver answering; controlled failure/recovery test passed
Discovery metering, four surfaces	Live, all returning no-store, meter recording in Production
Preview E2E gates	Passing on every pull request
Quality gate (typecheck, lint, tests, build, audit, migrations, worker)	Passing
Dependabot	Active — four update PRs already open
Revenue path readiness	One fault found — see section 3
Backup and restore	Tooling deployed; no rehearsal has been run
Alert escalation	Still a single destination, no acknowledgement

2. What shipped since the first assessment

2.1 Operational envelope

- **Reviewed migration path** — a dispatch-only workflow with a reviewer gate, an exact confirmation phrase, a schema-only snapshot, drift detection that stops the run if the live schema contains anything no migration describes, a dry-run mode, and post-apply readiness verification. It deliberately has no automatic revert.
- **Migration integrity in CI** — append-only enforcement, unique ordered UTC timestamps, and destructive DDL requiring a written justification. Runs on every pull request and again inside the migration workflow.
- **Restore rehearsal tooling** — verifies a restore an operator has performed into a scratch project and measures RTO and RPO with narrow definitions, so a quiet ledger is not mistaken for data loss.
- **Preview E2E gates** — the full integration suite now runs against each Preview deployment, so MCP discovery, gateway proxying, circuit-breaker behaviour, rate limiting, and audit export are checked before merge rather than after deploy.
- **Scheduled capacity trend** — a daily credit-free control-plane run against the newest ready Preview, recording the measured deployment and commit so a trend point is attributable.
- **Dependency governance** — Dependabot with the payment, database, telemetry, and model clients in their own review group, plus a production-dependency audit in CI. The two outstanding advisories were cleared.
- **Repository governance** — CODEOWNERS over the paths where a mistake is expensive, a security policy, and a README that states what the system is instead of create-next-app boilerplate.

2.2 Measurement

- **Revenue path readiness** — a private endpoint reporting which paid paths can actually transact, distinguishing a clean disabled state from a half-wired one, and naming any environment variable set under a near-miss name.
- **Agent discovery metering** — four surfaces counted against a seven-value client class derived per request and never stored. It answers the question the machine-economy strategy rests on: whether machines are finding the platform at all.
- **Orientation repaired** — the served `/11ms.txt` listed research claims and four mock endpoints but no commercial surface, so an agent reading the conventional orientation file was never led to the offers. It now carries the offer manifest, offers API, context note, agent card, and gateway contract.

3. Findings

Two of these were found by instrumentation added during this work, which is the point of having built it.

Sev.	Finding	Detail
P1	A revenue line is enabled but cannot transact	Prepaid MPS audit credits report enabled_incomplete: the checkout flag is on but STRIPE_MPS_CREDITS_WEBHOOK_SECRET is absent in Production. The configuration gate fails closed, so no money is at risk — but /mps/audit-access serves a live purchase page for a product that cannot be bought. Working notes show this variable pending since 17 July.
P1	Alerting still has no escalation	Release health failed for over an hour and the notification did not arrive, because the webhook receiver returned 503. The receiver is fixed, but the design is unchanged: one destination, no acknowledgement, no fallback. This is now a demonstrated failure rather than a theoretical one.
P1	Backup and restore remain unproven	The rehearsal tooling is deployed and the runbook is written, but no restore has been performed. Until one is, recovery time and data loss are unmeasured and the backup is unverified.
P2	Two documents contend for /llms.txt	A generated research claim index is served, while a 164-line curated orientation map in public/ is shadowed and unreachable. The curated file is the better orientation document. The conflict is a hard 500 in local development and invisible in Production only because the route is prerendered. Which document should own that URL is a content decision, so it was left open.
P2	The capacity trend will fail on its next run	The scheduled run resolves the newest Preview through the Vercel API, and the preview-capacity environment lacks VERCEL_TOKEN, VERCEL_TEAM_ID, and VERCEL_PROJECT_ID. Three settings, two minutes.
P3	The migration workflow has not been exercised	Both discovery migrations were applied by hand. A dry-run is queued and waiting on reviewer approval; it changes nothing and would confirm whether hand-applying left any drift.

4. What the numbers say so far

Discovery metering has been live for a matter of minutes, and the only traffic recorded is the verification requests made while confirming it works — nine requests, all classified `http_client`, zero from an agent runtime. Those nine should be discounted. Nothing can yet be concluded about agent demand, and the honest position is that the platform has no evidence either way until several weeks of data exist.

The measure that matters is not total requests but **agent-runtime requests against the offer manifest**: something deliberately reading the commercial terms. Crawler traffic is not demand. A sustained zero after a reasonable window is itself a finding, and a more useful one than building further on an untested assumption.

5. What remains

Step	Action	Why now
1	Set STRIPE_MPS_CREDITS_WEBHOOK_SECRET in Production and register the endpoint at /api/mps-credits/webhook, or take the purchase page down.	A live page advertises a product that cannot be bought.

Step	Action	Why now
2	Route operations alerts into a paging service with acknowledgement and one fallback recipient.	The failure mode has already occurred once and was silent.
3	Add the three Vercel settings to preview-capacity.	Prevents a daily failing workflow starting tomorrow.
4	Approve the queued migration dry-run and read its drift output.	Confirms hand-applied migrations left the schema consistent.
5	Perform one restore into a scratch project and run the verification script.	The last unmeasured catastrophic failure mode.
6	Decide which document owns /lms.txt and retire the other.	Resolves a latent conflict and an unreachable orientation map.
7	Review the four open Dependabot pull requests, including an ESLint major.	Dependency governance only works if the updates are triaged.
8	Wait several weeks, then read the discovery numbers.	The strategic question cannot be answered sooner.

Assessment. The gap identified in the first report has largely closed. The platform now has a reviewed path for schema changes, integration gates before merge rather than after deploy, dependency governance, and a measurement layer for both revenue readiness and agent discovery. What is left is smaller and more specific: one misconfigured revenue line, one missing escalation path, and one restore that has to actually be run. None of these require new engineering. The strategic question — whether autonomous agents want any of this — is now answerable with data rather than argument, which was the point, but it needs time to accumulate before anyone should act on it.